

CODE :-

```
import numpy as np
import numpy.random as ar

X = np.array([[0,0],[0,1],[1,0],[1,1]])
Y = np.array([[0],[1],[1],[0]])
lr=0.5
np.random.seed(0)

print("*****WEIGHTS*****")
W1 = ar.rand(2,2)
print("W1=", W1)

W2 = ar.rand(2,1)
print("W2=", W2)

def sigmoid(x):
    return 1 / (1 + np.exp(-x))

print("*****HIDDEN LAYER*****")
# Hidden layer
x = np.dot(X, W1)
h = sigmoid(x)
print("Output of hidden layer (h) =", h)

print("*****OUTPUT LAYER*****")
# Output layer
y = np.dot(h, W2)
o = sigmoid(y)
print("Output of output layer (o) =", o)

def sigmoid_d(x):
    return x*(1-x)

for i in range (3000):
    x = np.dot(X, W1)
    h = sigmoid(x)
    y = np.dot(h, W2)
    o = sigmoid(y)

    e=Y-o
    d_output=e*sigmoid_d(o)
    d_hidden=(np.dot(d_output,W2.T))*sigmoid_d(h)

    if(i%500==0):
        print("epoch:",i)
        print("MSE=",np.mean(e**2))
        print("d_output=",d_output)
        print("d_hidden=",d_hidden)
```

```
W2=W2+np.dot(h.T,d_output)*lr
W1=W1+np.dot(X.T,d_hidden)*lr
```

```
print("*****FINAL OUTPUT*****")
o = [[1 if val > 0.5 else 0] for [val] in o]
o = np.array(o)
print("o=",o)
print("*****FINAL WEIGHTS*****")
print("W2=",W2)
print("W1=",W1)
```

OUTPUT :-

```
*****WEIGHTS*****
W1= [[0.5488135  0.71518937]
      [0.60276338 0.54488318]]
W2= [[0.4236548 ]
      [0.64589411]]
*****HIDDEN LAYER*****
Output of hidden layer (h) = [[0.5      0.5      ]
      [0.64628827 0.63294764]
      [0.63386027 0.6715468 ]
      [0.75979882 0.7790386 ]]
*****OUTPUT LAYER*****
Output of output layer (o) = [[0.63059599]
      [0.66432327]
      [0.66869443]
      [0.69531126]]

epoch: 0
MSE= 0.2758878242324435
d_output= [[-0.14689399]
      [ 0.07485519]
      [ 0.07339816]
      [-0.14730413]]
d_hidden= [[-0.01555809 -0.02371949]
      [ 0.00724953  0.01123257]
      [ 0.00721668  0.01045674]
      [-0.0113894  -0.01637765]]

epoch: 500
MSE= 0.24658452289279348
d_output= [[-0.12319348]
      [ 0.12187354]
      [ 0.12070234]
```

[-0.13011075]]
d_hidden= [[0.01864481 -0.01688895]
[-0.01765177 0.01234278]
[-0.01770585 0.0117107]
[0.01700556 -0.00572444]]

epoch: 1000
MSE= 0.17299556595921134
d_output= [[-0.08870866]
[0.09334915]
[0.09306153]
[-0.12327271]]
d_hidden= [[0.09733149 -0.0751458]
[-0.0929406 0.00785917]
[-0.09277575 0.00788525]
[0.09352039 -0.00028747]]

epoch: 1500
MSE= 0.11175605649257249
d_output= [[-0.05393769]
[0.06934384]
[0.06934187]
[-0.09960916]]
d_hidden= [[1.14911966e-01 -8.83938584e-02]
[-1.29719210e-01 3.34998751e-03]
[-1.29718716e-01 3.35702558e-03]
[1.29986372e-01 -3.66113207e-05]]

epoch: 2000
MSE= 0.07902763587631684
d_output= [[-0.03539015]
[0.05355204]
[0.05355163]
[-0.08046583]]
d_hidden= [[1.02745051e-01 -7.95035520e-02]
[-1.34069599e-01 1.91045252e-03]
[-1.34069525e-01 1.91264344e-03]
[1.34220252e-01 -1.15930378e-05]]

epoch: 2500
MSE= 0.05976746480745469
d_output= [[-0.02502768]
[0.04293533]
[0.04293518]
[-0.06635506]]

```
d_hidden= [[ 8.75799267e-02 -6.80957972e-02]
[-1.28092701e-01  1.26437691e-03]
[-1.28092679e-01  1.26536951e-03]
[ 1.28189506e-01 -5.34927851e-06]]
```

```
*****FINAL OUTPUT*****
```

```
o= [[0]
[1]
[1]
[0]]
```

```
*****FINAL WEIGHTS*****
```

```
W2= [[-15.91089997]
[ 12.41536424]]
W1= [[0.82966612 6.17375412]
[0.82967122 6.1743614 ]]
```